

PITN: Physics-Informed Temporal Networks for Cuffless Blood Pressure Estimation

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Abstract—Estimating blood pressure (BP) plays a crucial role in healthcare. Traditionally, BP has been measured using a cuff, which is unsuitable for continuous BP monitoring. Recent advancements in cuffless wearable devices, such as graphene electronic tattoos and PPG sensors provide viable alternatives. However, existing BP estimation methods often overlook the importance of temporal modeling, despite the multi-periodicity and temporal dependencies inherent in cuffless BP signals. Moreover, continuous BP monitoring requires personalized modeling, which is further challenged by data scarcity. To address these two challenges, we introduce a novel Physics-Informed Temporal Network (PITN) with adversarial contrastive learning to enable precise BP estimation with very limited data for three different modalities (i.e., bioimpedance, PPG, millimeter wave). Specifically, we first introduce a novel Physics-Informed Temporal Network for investigating BP dynamics' multi-periodicity for cardiovascular cycle modeling and temporal variation. We then apply adversarial training to generate extra physiological time series data, improving PITN's robustness in the face of sparse subject-specific training data. Furthermore, we utilize contrastive learning to capture the discriminative variations of cardiovascular physiologic phenomena. This approach aggregates physiological signals with similar blood pressure values in latent space while separating clusters of samples with dissimilar blood pressure values. Experiments on three widely-adopted datasets with different modalities demonstrate the superiority and effectiveness of the proposed methods over previous state-of-the-art approaches.

Index Terms—Cuffless blood pressure (BP) estimation, physics-informed neural network, multimodal sensing, adversarial learning.

I. INTRODUCTION

CONTINUOUS monitoring of vital physiological signs such as heart rate (HR), blood pressure (BP), and respiration rate (RR), is essential and has become popular on smart

wearables. The blood circulatory system can be considered as a whole-body interconnecting organ, hence, blood pressure (BP) is one of the most crucial indicators of cardiovascular health [1]. Two common irregularities in BP are hypertension (high blood pressure) and hypotension (low blood pressure), which can easily be overlooked despite indicating certain cardiovascular diseases such as stroke or chronic kidney issues. Therefore, accurate blood pressure estimation holds significant importance in monitoring everyday human health. To enable continuous monitoring of BP values, various wearable devices have been integrated [2] to mine valuable information for health analysis, such as graphene electronic tattoos and multiple sensors. These devices enable precise BP estimation using diverse multimodal signals, including bioimpedance, millimeter wave, and photoplethysmography (PPG). This casts a problem of how these multi-sensor data should be analyzed to predict the subject's health information. They significantly enhance user experience and contribute to the advancement of digital health.

Traditionally, blood pressure is measured using a cuff to compress the arm, which causes an unpleasant user experience [4]. As a result, cuffless BP estimation has gained significant development in recent years to improve user comfort. A lot of ambulatory BP monitoring platforms provide multiple solutions for continuous cuffless BP estimation in non-clinical (ambulatory) settings using wearable devices like bioimpedance [5], [6], [7] or millimeter-wave sensor [3]. Considering that these electrical signals are frequently a multi-variable measurement utilized to predict physiological states, recent developments in AI algorithms, especially deep learning models, can provide great opportunities for cuffless BP estimation. This involves inferring BP using data from cuffless wearable devices (e.g., Blumio sensor, graphene electronic tattoos), by modeling the complex inherent input-output temporal relationships in physiological systems. However, current BP estimation methods fall short in temporal modeling. Furthermore, continuous monitoring of specific subjects requires personal modeling, which usually encounters problems such as limited ground truth data. To address these challenges, there is a pressing need to develop advanced models for physiological time series data that can achieve accurate BP estimation with reduced reliance on ground truth data.

Extensive efforts [5], [6], [8], [9], [10] have been devoted to cuffless BP estimation, which utilizes various methods for modeling, such as Finite Element Analysis (FEA) [5], Bidirectional Long Short Term Memory (BiLSTM)-based Hybrid Neural

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The code is available at <https://github.com/Zest86/ACL-PITN>.

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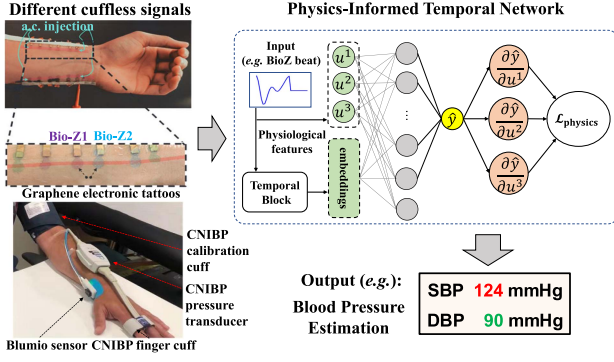


Fig. 1. Illustration of the cuffless blood pressure estimation task and our proposed PITN with bioimpedance signals in graphene-HGCPT dataset (left top) [1], PPG and millimeter-wave signal in Blumio dataset (left bottom) [3].

Network (HNN) [8] and Autoencoder [9], etc. These models work in a pure data-driven framework, without utilizing domain knowledge in biological systems, which makes them effective in numerous training data situations but fails in personalized modeling. Recently, the emerging physics-informed neural networks (PINNs) [11], which leverage underlying physics laws structured by generalized partial differential equations (PDEs), have proven their superiority in related fields such as structural mechanics [12], mechanical systems [13], and biomedical informatics [14]. Compared with the above-mentioned pure data-driven BP estimation approaches, PINN utilizes physiological prior to formalizing additional physics residual for constraint on model training, which enhances BP estimation with reduced labels for training [6].

While utilizing underlying physics laws, PINNs fall short in temporal modeling [15]. In fact, beat-to-beat BP estimation is a continuous time series data mining task, where input time series waveform contains rich temporal information. There are some Transformer-based methods that utilize the long-term modeling ability for cuffless blood pressure estimation [16]. Specifically, Transformer-based models adopt the attention mechanism or its variants to capture the pair-wise temporal dependencies among time points [17]. However, it is still challenging for the attention mechanism to directly identify reliable dependencies from scattered time points, as these temporal dependencies can be deeply obscured by intricate temporal patterns [18]. To bridge the gap between PINN and temporal modeling, we propose a novel Physics-Informed Temporal Network (PITN) for cuffless blood pressure estimation, which enhances PINN's ability in temporal data mining through a novel temporal block for personal cardiovascular cycle modeling, as shown in Fig. 1. The proposed temporal block enables temporal variation modeling through transformation of 1D time series to 2D space and is suitable for personalized cardiovascular cycle modeling.

For data scarcity, one promising way is adversarial training, which is used to improve the model's robustness. Several recent works show that adversarial examples can be used to improve model performance in computer vision (CV) [19], natural language processing [20]. These previous works make it promising to use adversarial samples for data augmentation in time series analysis. Unfortunately, limited work adopts such a similar

adversarial training strategy in medical time series data, which is our main focus of data augmentation methods in this work.

In addition, contrastive representation learning has been shown as a promising manner to mitigate data scarcity [21]. However, extending contrastive learning paradigms to time series presents significant challenges, especially in the health domain with unique characteristics (e.g., low frequency, and high sensitivity [22]). Although, some previous work adopt self-supervised contrastive learning to mitigate the challenge of label scarcity in medical time series [21], [23]. These self-supervised methods require a large amount of data for pre-training, which is an emergent weakness. Besides, Sel et al. [6] point out that for deep learning-based BP estimation models, as the training data increases, personalized correlation fluctuates.

For this part, to the best of our knowledge, there is no related work focusing on improving this insufficiency. In this work, we utilize contrastive learning to address this problem by adding soft constrain on latent embedding, to enable the deep model to learn from physiological signals with similar BP values. Through using generated adversarial examples, our proposed method can offer extra alternatives for the deep model to capture the discriminative variations of cardiovascular physiologic phenomena.

The main advantages of our proposed framework are three-fold. First, we propose a novel Physics-Informed Temporal Network (PITN) with the newly designed temporal block to capture intra-periodic temporal features. Second, we further enhance the PITN by incorporating adversarial training, which generates extra adversarial samples to ensure sufficient robustness. These perturbed samples also serve as data augmentation to boost model training. Last but not least, the proposed contrastive learning leverages adversarial samples to better capture the discriminative BP dynamics. Our contributions can be summarized as follows:

- We propose a novel Physics-Informed Temporal Network (PITN) for cuffless blood pressure estimation by enhancing the PINN with the newly designed temporal blocks, which model precise personal temporal characteristics and can be applied to different modalities.
- We propose a novel adversarial training method to augment physiological time series data, which leverages physiological priors during the generation.
- We design a contrastive learning approach incorporated with adversarial training to capture blood pressure discriminative variations.
- Extensive experiments are carried out on graphene-HGCPT, ring-CPT, and Blumio datasets, and results show our methods' superiority in different modal physiological time series modeling using minimal ground truth data.

II. RELATED WORK

A. Blood Pressure Monitoring

Blood Pressure (BP) is a key cardiovascular parameter commonly used by clinicians to evaluate cardiac and circulatory health [24]. Traditionally, BP is measured using cuff-based (auscultatory or oscillometric) digital BP devices, which compress the user's arm and often result in an unpleasant user

experience [25]. With the increasing use of cuffless sensors for health monitoring [26], recent work tends to explore different modal cuffless BP measurement methods (e.g., bioimpedance, millimeter wave, PPG), which use wearable devices to capture pulse activities and infer BP using learning-based methods, like BiLSTM-based HNN [8], and Transformer [4], [9]. Despite these advancements, most learning-based approaches still rely on large amounts of invasive or cuff-based BP measurements as ground truth labels, limiting their practicality in real-world applications. Recently, physics-informed neural networks (PINNs) have been proposed to incorporate physical residuals into BP estimation, leading to improved generalization. These physical residuals can be computed in an unsupervised manner, mitigating the data scarcity problem. However, PINN-based methods primarily focus on enforcing consistency with physical constraints and often overlook the temporal dependencies inherent in BP waveforms, leading to suboptimal performance in precise physiological modeling. Particularly, beat-to-beat BP estimation and learning algorithms for continuous BP wave function extraction can be viewed as a time series modeling task, which is of significant importance in data mining and has attracted increasing attention in multiple areas [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]. Numerous models have been developed to analyze these beat-to-beat time series signals [38], [39], which includes time domain LSTM-based methods [38], a mixture of time and frequency domain human designed features-based approaches [39]. To address the limitations of existing methods, we propose a novel personalized modeling approach that integrates physical residuals with intra-periodic feature extraction from BP waveforms. This time-series modeling framework enhances the temporal consistency of BP estimation, enabling more accurate and personalized predictions.

B. Adversarial Training

Numerous studies have provided supporting evidence for the proposition that training with adversarial examples can effectively improve the capabilities of models [19], [20], [40], [41]. Compared to clean examples, adversarial ones contribute to a more effective alignment of network representations with salient data characteristics and human perception [42]. Furthermore, models trained with adversarial examples demonstrate heightened robustness to high-frequency noise [43]. Besides, previous work [44] also demonstrates that by automatically generating adversarial examples, adversarial training will not decrease model performance but benefit both generalization and robustness of models. He et al. [45] use adversarial training to generate multivariate time-series samples for forecasting. Especially for physics-informed neural networks (PINNs), of which the robustness can be enhanced by fine-tuning the model with adversarial samples [46]. To the best of our knowledge, we are the first to adopt such an adversarial training method as data augmentation in BP estimation.

C. Contrastive Learning

Contrastive learning has shown remarkable success in deep learning tasks [47], [48]. Existing techniques like SimCLR [49]

and SimCSE [50] have demonstrated superior performance in generating semantically rich image embeddings without the need for labeled data. These methods help pull instances together with the same labels in the latent space, while simultaneously pushing clusters of samples from different classes apart. However, contrastive learning is rarely applied to time series analysis [51]. Several common image augmentations, including color changes or rotation may not be as relevant to time series data, especially in medical time series data mining. To address this insufficiency, we introduce contrastive learning to enhance the performance by capturing discriminative variations in cardiovascular dynamics.

III. PRELIMINARIES

A. Problem Definition

Given the input time series signal $\mathbf{X} \in \mathbb{R}^{N \times T \times C}$, the goal of cuffless blood pressure is to estimate blood pressure value y (e.g., systolic BP (SBP) or diastolic BP (DBP)). Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$ be the continuous bioimpedance (BioZ) signal with N segments, length- T and C channels, for each segment $\mathbf{x}_i \in \mathbb{R}^{T \times C}$, we extract physiological feature $\mathbf{u}_i \in \mathbb{R}^M$ denoted as the M dimensional vector. The neural network can be formulated as $\hat{y} = f(\mathbf{x}, \mathbf{u}, \theta)$ with input $\mathbf{x}$, $\mathbf{u}$, parameter θ , and the predicted blood pressure $\hat{y}$.

B. Physics-Informed Neural Networks

PINNs are trained to solve regular supervised tasks while adopting several given laws of physics described by partial differential equations (PDEs) as the network residual. Inspired by [6], we build physics constraints with Taylor's approximation for certain gradually changing cardiovascular phenomena (e.g., establishing the relationship between physiological features extracted from bioimpedance sensor measurements and BP). Here we can define a polynomial with Taylor's approximation around i -th segment as follows:

$$\tilde{f}_i(\mathbf{x}, \mathbf{u}, \theta) = f(\mathbf{x}_i, \mathbf{u}_i, \theta) + \nabla_{\mathbf{u}_i} f(\mathbf{x}_i, \mathbf{u}_i, \theta)^T (\mathbf{u} - \mathbf{u}_i), \quad (1)$$

where $\tilde{f}_i(\mathbf{x}, \mathbf{u}, \theta)$ represents the above Taylor polynomial approximated based on i -th segment. The first-order Jacobian matrix can be calculated discretely as $\sum_{k=1}^M \frac{\partial f}{\partial \mathbf{u}^k} (\mathbf{u}^k - \mathbf{u}_i^k)$.

We can calculate a physics-based loss for the neural network using the residual value evaluated at $(i+1)$ -th segment using Taylor's approximation around i -th segment as in Eq. (2):

$$\mathcal{L}_{\text{physics}} = \frac{1}{(N-1)} \sum_{i=1}^{N-1} \left(\tilde{f}_i(\mathbf{x}_{i+1}, \mathbf{u}_{i+1}, \theta) - f(\mathbf{x}_{i+1}, \mathbf{u}_{i+1}, \theta) \right)^2, \quad (2)$$

where N is the total number of segments. Given that $\mathcal{L}_{\text{physics}}$ is calculated in an unsupervised way, the physics loss can be calculated for any given input sequence.

As for physiological feature $\mathbf{u}$, we denote the amplitude change as u^1 , which serves as a proxy for the extent of arterial expansion. The second feature is represented by u^2 , which captures the inverse of the relative time difference between the forward-traveling (*i.e.*, systolic) wave and the reflection wave,

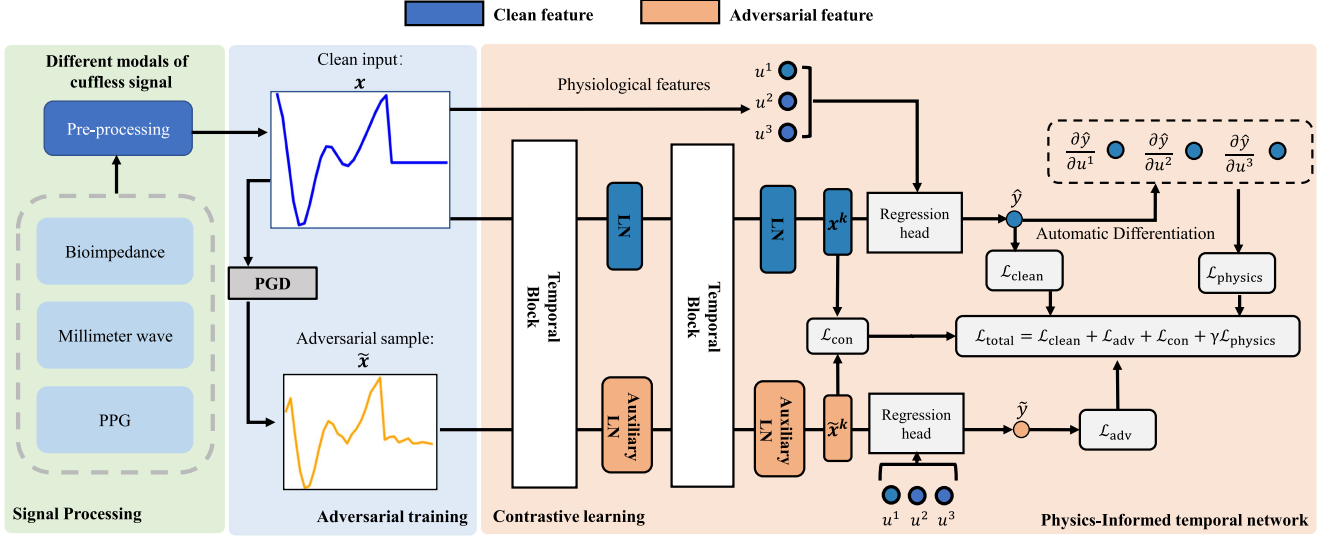


Fig. 2. Overall framework of physics-informed temporal network with adversarial contrastive learning for different modal cuffless blood pressure estimation. Our framework mainly contains two data flows: the upper part is about clean signals, which generate predicted values and differentiation for Taylor's approximation. The bottom part is adversarial samples generated by PGD accordingly. Notably, we introduce the adversarial training method in the PGD and construct the contrastive module to enable the model's ability in limited physiological data training. In the inference phase, the auxiliary LNs are not activated for different data distributions between clean and adversarial samples.

providing an estimate of the pulse wave velocity (PWV). The third feature denoted as u^3 , which corresponds to beat-to-beat heart rate. These three physiological features can be formulated as follows:

$$u^1 = \Delta Z_{\max} - \Delta Z_{\min}, u^2 = \frac{1}{(t_F - t_B)}, u^3 = \frac{60}{t_J - t_A}, \quad (3)$$

where $\Delta Z_{\max}, \Delta Z_{\min}$ denotes the maximum and minimum variation in the (inverted) bioimpedance signal, (t_F, t_B) refer to the trough point in the derivative signal and the zero-crossing point in the derivative signal transitioning to the ascent, and (t_J, t_A) denote the peak point in the derivative signal, the zero-crossing point in the derivative signal transitioning to the descent, respectively.

IV. PROPOSED APPROACH

A. Overview

As illustrated in Fig. 2, our framework comprises three key components: 1) Physics-Informed Temporal Network (PITN), 2) Adversarial Training, and 3) Contrastive Learning. In the PITN, newly designed temporal blocks are employed to extract personal physiological features, and Projected Gradient Descent (PGD) is tailored to generate adversarial samples for model training augmentation. These adversarial samples, along with clean samples, are disentangled using distinct layer normalization (LN) layers. Additionally, we adopt the contrastive learning loss by comparing clean and adversarial samples, based on their true blood pressure (BP) values.

The whole process is as follows. We first pre-process different modal signals (e.g. bioimpedance signal) to obtain a clean input $\mathbf{x}$. Then adversarial sample $\tilde{\mathbf{x}}$ is generated using Projected Gradient Descent (PGD). Both clean and adversarial inputs are fed

through the temporal block for feature extraction and the output estimated BPs are denoted as $\hat{y}$ and $\tilde{y}$, respectively.

B. Physics-Informed Temporal Networks

The BP signal is a continuously varying dynamic process, and its instantaneous values may be affected by noise or short-term fluctuations. As a result, information from a single time point is insufficient for accurate BP estimation. Capturing the temporal dynamics of BP variations is crucial for improving estimation accuracy. Physics-informed neural networks (PINNs) traditionally use coordinates as input while overlooking potential temporal relationships among these inputs. This omission disregards the crucial inherent temporal dependencies present in practical physical systems, leading to a failure in globally propagating initial condition constraints and capturing accurate solutions across diverse scenarios [15]. In this paper, we introduce a novel temporal modeling block to enhance the capability of PINNs in modeling time-series medical data by extracting complex temporal variations from transformed 2D tensors using a multi-scale Inception block.

As illustrated in Fig. 3, we organize the temporal block in a residual way [52]. Specifically, for the length- T 1D input signal $\mathbf{x} \in \mathbb{R}^{T \times C}$, we project the raw inputs into the deep features $\mathbf{x}^0 \in \mathbb{R}^{d_{\text{model}}}$ through the initial embedding layer, represented as $\mathbf{x}^0 = \text{Embed}(\mathbf{x})$. For the k -th layer of temporal block, the input $\mathbf{x}^{k-1} \in \mathbb{R}^{d_{\text{model}}}$ is processed as:

$$\mathbf{x}^k = \text{TemporalBlock}(\mathbf{x}^{k-1}) + \mathbf{x}^{k-1}. \quad (4)$$

Each temporal block is designed to capture both intra-period personal characteristics of physiological signals, as shown in Fig. 3. This process includes operations such as $\text{Period}(\cdot)$, two $\text{Reshape}(\cdot)$ functions, a 2D inception convolution $\text{Inception}(\cdot)$ and residual addition. Initially, the 1D signal is extended to a 2D

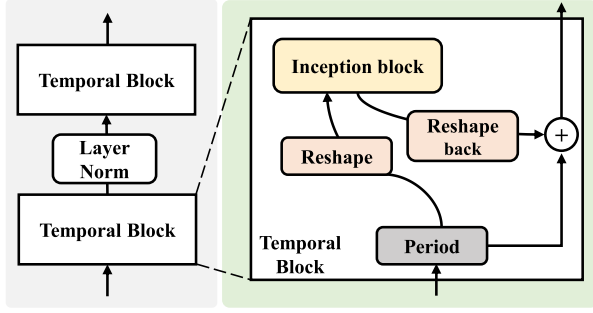


Fig. 3. Illustration of the proposed temporal block, which is designed to capture personalized cardiovascular cycles from the 1D signal by transforming it into a 2D tensor, followed by feature extraction using an inception block. The temporal block is stacked in a residual manner.

shape by calculating:

$$A = \text{Avg}(\text{Amp}(\text{FFT}(\mathbf{x}))), f = \text{argmax}(A), p = \left\lceil \frac{T}{f} \right\rceil, \quad (5)$$

where $\text{FFT}(\cdot)$ and $\text{Amp}(\cdot)$ denote the Fast Fourier Transform and the calculation of amplitude values, respectively. A represents the calculated amplitude for each frequency, averaged by $\text{Avg}(\cdot)$. The personal cardiovascular cycle is identified by detecting the periodic basis function as described in Eq. (5). For computational efficiency and avoidance of noise [53], only the most significant frequency is selected for personalized modeling. The entire process described in Eq. (5) can be summarized as $(A, f, p) = \text{Period}(\mathbf{x})$.

For processing an extended 2D signal, the temporal block is formulated as follows:

$$\mathbf{X}_{2D}^k = \text{Reshape}_{p,f}(\text{Padding}(\mathbf{x}^k)), \quad (6)$$

where $\text{Padding}(\cdot)$ extends the time series by zeros along the temporal dimension to ensure compatibility with $\text{Reshape}(\cdot)$. For 2D signal encoding, $\text{Inception}(\cdot)$ is employed, based on the InceptionNet [54], which utilizes multi-scale 2D kernels, formulated as:

$$\hat{\mathbf{X}}_{2D}^k = \text{Inception}(\mathbf{X}_{2D}^k), \quad (7)$$

where $\mathbf{X}_{2D}^k \in \mathbb{R}^{p \times f \times d_{\text{model}}}$ is the transformed 2D tensor. The learned 2D representation $\hat{\mathbf{X}}_{2D}^k$ is then transformed back to 1D space $\mathbf{x}^k \in \mathbb{R}^{d_{\text{model}}}$ for subsequent stacked temporal blocks. After obtaining the temporal embeddings $\mathbf{x}^k$, the physiological features $\mathbf{u}$ are concatenated for regression. The estimated blood pressure value is then computed as follows:

$$\hat{y} = \text{RegHead}([\mathbf{x}^k \parallel \mathbf{u}]), \quad (8)$$

where $\text{RegHead}(\cdot)$ represents a fully connected layer, and $\parallel$ denotes concatenation. The embedding $\mathbf{x}^k$ encapsulates rich temporal features from the sensor signal, which complements the PINN model. By integrating these two sets of features, precise blood pressure measurements can be achieved.

C. Adversarial Training

Due to the limited availability of labeled physiological time series data, the results of current methods are still unsatisfactory.

Algorithm 1: Physiological time series data generated by PGD.

Input: Clean BioZ data $\mathbf{x}$, problem domain Ω , proposed model $f(\mathbf{x}, \mathbf{u}, \theta)$.

Parameter: Adversarial perturbation $\Delta \sim \mathcal{N}(0, \sigma^2)$, number of iteration steps I , and the step size of each iteration η .

Output: Adversarial samples $\tilde{\mathbf{x}}$.

```

1: for  $t = 1 : I$  do
2:    $\tilde{\mathbf{x}} = \mathbf{x} + \Delta$  // Perturbation
3:    $\tilde{\mathbf{x}} = \text{clip}_{\Omega}(\tilde{\mathbf{x}})$ 
4:    $\Delta = \eta \times \text{sign}(\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{u}, \theta))$  // Gradient ascent
5:    $\tilde{\mathbf{x}} = \tilde{\mathbf{x}} + \Delta$ 
6:    $\tilde{\mathbf{x}} = \text{clip}_{\Omega}(\tilde{\mathbf{x}})$ 
7: end for
8: return  $\tilde{\mathbf{x}}$ 

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To address this issue, we enable blood pressure estimation using adversarial training by generating additional samples, while preserving the performance on clean data [40].

For estimating BP with the function $f(\mathbf{x}, \mathbf{u}, \theta)$, we formulate the adversarial training as an optimization problem:

$$\begin{aligned} \max_{\Delta} \quad & \mathcal{L}(f(\mathbf{x} + \Delta, \mathbf{u}, \theta), f(\mathbf{x}, \mathbf{u}, \theta)) \\ \text{s.t.} \quad & \|\Delta\|_{\infty} \leq \epsilon, \end{aligned} \quad (9)$$

where $\mathcal{L}(\cdot)$ represents the mean square error between $f(\mathbf{x} + \Delta, \mathbf{u}, \theta)$ and $f(\mathbf{x}, \mathbf{u}, \theta)$, and ϵ is the threshold for the maximum allowable adversarial perturbation. The perturbation is initialized drawn from a uniform distribution $\Delta \sim \mathcal{N}(0, \sigma^2)$. Subsequently, the adversarial perturbation is updated as $\Delta = \eta \times \text{sign} \nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{u}, \theta)$, where sign is the element-wise sign function, producing -1 for negative values of the Jacobin matrix and $+1$ otherwise. η denotes the step size of each iteration.

In our proposed blood pressure estimation framework, we further regularize the perturbed signal by truncating it within the problem domain Ω :

$$\tilde{\mathbf{x}} = \text{clip}_{\Omega}(\mathbf{x} + \Delta), \quad (10)$$

where the problem domain is defined by the element-wise maximum and minimum values of the input, $\Omega = [\min(\mathbf{x}), \max(\mathbf{x})]$ for all training data. We adopt the Projected Gradient Descent (PGD) [55] in time series adversarial samples generating, as shown in Algorithm 1. During initialization, the clean BioZ data $\mathbf{x}$ is perturbed with randomly sampled variables Δ from the uniform distribution. The $\text{clip}_{\Omega}(\cdot)$ means clipping samples according to the problem domain Ω . During the projected gradient descent process, gradient ascent is applied to maximize the residual, and the perturbed sample is subsequently truncated again.

D. Contrastive Learning

Contrastive learning shows potential in mitigating data scarcity, but its application in time series is limited. Recently, contrastive learning has achieved success in self-supervised

learning [56], by introducing additional constraints in the latent space to regularize embedding with the same label. Specifically, in classification tasks, similar data pairs (referred to as positive pairs) are selected for the same class, while all other pairs are treated as negative ones. However, applying contrastive learning to regression tasks remains non-trivial. In this part, we first introduce contrastive learning into the BP regression issue. Here, we define the i -th and j -th samples as a positive pair based on the following criterion:

$$y_{\text{shift}} \geq |y_i - y_j|, \quad (11)$$

where y_{shift} is a threshold parameter. By imposing this soft constraint, which labels inputs with similar blood pressure (BP) values, we aim to enhance the model's ability to learn cardiovascular dynamics from a single pulse wave. The contrastive learning loss is computed between clean and adversarial samples. Inspired by [57], we formalize the contrastive learning loss as follows:

$$\mathcal{L}_{\text{con}} = \sum_{i \in S} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{x}_i^k \cdot \mathbf{x}_p^k / \tau)}{\sum_{a \in A(i)} \exp(\mathbf{x}_i^k \cdot \mathbf{x}_a^k / \tau)}, \quad (12)$$

where $i \in S$ is the index of an arbitrary training sample, and index p denotes an adversarial sample which has $|y_p - y_i| < y_{\text{shift}}$. Index p is positive and index a is negative. $P(i)$ and $A(i)$ are sets of positive samples and negative samples, respectively. The embeddings $\mathbf{x}_p^k$ and $\mathbf{x}_a^k$ correspond to positive/negative pairs, respectively, after k layers of Temporal blocks for the input $\mathbf{x}_i$. The symbol $\cdot$ represents the inner (dot) product, and τ is a scalar temperature parameter.

E. Training and Inference

1) *Training*: We formally integrate the Projected Gradient Descent (PGD) method into the regression framework, as outlined in Algorithm 1. For each clean input, we initiate the process by using PGD to generate its corresponding adversarial counterpart. Subsequently, both the clean data and its adversarial counterpart are fed into the same network but with different normalization layers (LNs) inspired by [40]. Specifically, the primary LNs are applied to the clean input, while the auxiliary LNs are used for the adversarial counterpart. The overall loss is then minimized with respect to the network parameters through gradient updates. It is important to note that, aside from the LN layers, all other layers are optimized concurrently for both clean inputs and their adversarial counterparts.

During the training phase, both clean and adversarial samples are input into the network to obtain blood pressure predictions. The regression loss $\mathcal{L}_{\text{clean}}$ and $\mathcal{L}_{\text{adv}}$, are calculated as follows:

$$\begin{aligned} \mathcal{L}_{\text{clean}} &= \frac{1}{S} \sum_{i=1}^S (\hat{y}_i - y_i)^2, \\ \mathcal{L}_{\text{adv}} &= \frac{1}{S} \sum_{i=1}^S (\tilde{y}_i - y_i)^2, \end{aligned} \quad (13)$$

where S denotes the total number of training samples, and $\hat{y}_i$ and $\tilde{y}_i$ correspond to the network outputs for the i -th clean example

$\mathbf{x}_i$ and adversarial example $\tilde{\mathbf{x}}_i$, respectively. The overall loss is computed by summing the regression loss from Eq. (13), the physical loss from Eq. (2), and the contrastive loss from Eq. (12). The total loss can be formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{con}} + \gamma \mathcal{L}_{\text{physics}}, \quad (14)$$

where γ is a hyper-parameter which is set to 1 for balancing the magnitude of multiple losses. In minimal training criteria, there is no validation set. Therefore, we adopt the early stopping criterion from [6], which halts training if the supervised loss $\mathcal{L}_{\text{supervised}} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{con}}$ drops below 0.01 or if the maximum number of training epochs is reached.

2) *Inference*: Given a test bioimpedance segment $\mathbf{x} \in \mathbb{R}^{T \times C}$ representing an entire cardiac cycle, along with physiological features $\mathbf{u} \in \mathbb{R}^M$, the network output $\hat{y} = f(\mathbf{x}, \mathbf{u}, \theta)$ is obtained using primary layer normalizations (LNs). The output corresponds to the blood pressure (BP) value (e.g., systolic BP or diastolic BP).

V. EXPERIMENTS

In this section, we comprehensively evaluate the performance of our proposed framework on three public benchmark datasets, containing various wearable signals for cuffless blood pressure estimation. The evaluation aims at estimating systolic BP (SBP) and diastolic BP (DBP).

A. Datasets

We conduct our experiments on graphene-HGCPT [1], ring-CPT [5] and Blumio [3] datasets. The first two datasets contain raw time series data obtained from a wearable bioimpedance sensor, with corresponding reference BP values acquired using a medical-grade finger cuff (Finapres NOVA). The Blumio dataset includes data from 115 subjects (aged range 20-67 years), collected using several types of wearable sensors: PPG, applanation tonometry, and the Blumio millimeter-wave radar.

1) *Graphene-HGCPT Dataset [1]*: This BioZ signal based dataset includes data from six participants who undergo multiple sessions of a blood pressure elevation routine, which involves hand grip (HG) exercise followed by a cold pressor test (CPT) and recovery. Participants wear bioimpedance sensors that utilize graphene e-tattoos placed on their wrists along the radial artery. They also wear a silver electrode-based wristband at different positions. The evaluation takes over 24,829 samples (after post-processing), covering a wide range of BP values.

2) *Ring-CPT Dataset [5]*: This dataset includes data from five participants who underwent multiple sessions of CPT and recovery. Bioimpedance data were collected using a ring-worn BioZ sensor placed on the participants' fingers. During the bioimpedance signal recording, a cuff-based brachial BP measurement, and a finger cuff (Finapres NOVA) were attached to each participant for time calibration. The overall evaluation is based on a total of 6,544 samples from the ring-CPT.

3) *Blumio Dataset [3]*: This multimodal dataset includes data from 115 subjects using multiple wearable sensors. In our experiments, we select a subset of 30 subjects, aged 21 to 50 years (15 males and 15 females), and test our framework

on PPG and millimeter-wave signals to evaluate our methods across different modalities for cuffless BP estimation. For data processing, we employ the same procedure as for the ring-CPT and graphene-HGCPT datasets.

We conduct experiments across three datasets utilizing three modalities (i.e. BioZ, PPG, and mmWave) to demonstrate the versatility of our models. Each modality captures distinct aspects of blood pulsations, with its own strengths and limitations. For example, bioimpedance measures impedance changes in deep skin and underlying tissues by injecting low-power alternating current, reflecting blood volume changes in larger vessels. In contrast, PPG detects blood volume changes in the skin's microvascular layer using light and photodetectors [58]. Thus PPG is more sensitive to changes in superficial tissues but is prone to external noise and artifacts. Millimeter-wave sensors, uses radar to detect small surface vibrations caused by blood pulsation. These sensors capture surface-level micro-vibration, which is non-contact. However, mmWave sensors are limited by resolution and signal-to-noise ratio [59]. Our proposed methods are well-suited to these three modalities and demonstrate superior results.

B. Experimental Settings

Our experiments adhere to a minimal training criterion [6], where the labeled data is divided into S bins. Here, S represents the difference between the maximum and minimum BP values, calculated separately for SBP and DBP. This binning process partitions the dataset into distinct bins with a bin width of 1mmHg, from which one data point is randomly selected from each bin to form the initial training set. To ensure a fair comparison, we adopt the same data partitioning strategy, training configuration, and experimental conditions.

Due to the inaccessibility of the preprocessing in paper [6], we reproduce the process ourselves. We apply a consistent preprocessing method across all three datasets, extracting beat-to-beat signals from a single channel to ensure that our approach remains universal across different datasets and modalities. This approach differs slightly from the method used in [6], which integrates multiple channels in the graphene-HGCPT dataset. For each of the three datasets, separate models are trained for each subject to achieve precise, personalized modeling. Each model receives a varying number of labeled training points according to the minimal training criterion. Subsequently, we evaluate the performance of each model against reference blood pressure (BP) values using a test set, which includes BP values not used in the training process.

For preprocessing, following [6], we synchronize and acquire bioimpedance and blood pressure data (from Finapres NOVA) simultaneously for each dataset. The data is then divided into intervals corresponding to individual heartbeats, identified by automatically detected fiducial points. To isolate the blood volumetric changes, which follow the heart rate, from the raw bioimpedance data stream, we apply a second-order low-pass Butterworth filter with a 6 Hz cutoff frequency. This filtering process removes noise outside the desired frequency range while still enabling the measurement of extreme heart rates. Next, we

identify the consecutive maximum slope points of the systolic wave by locating the peak amplitudes of the first inverted derivative of the signal. We ensure that the peak detection algorithm selects the most prominent peaks that align with the maximum slope points in the original signal, ensuring plausible periodicity around the heart rate. Once the maximum slope points are identified, the zero-crossing points prior to these maximum slope points in the first derivative signal are marked as the start of the current cardiac cycle and the end of the previous one.

C. Baselines

We compare our model against several baseline models and state-of-the-art approaches, including methods specifically designed for cuffless blood pressure estimation as Hybrid-LSTM [60], Physics-informed neural networks (PINN) [6] and ResNet1D [4]. Additionally, we evaluate our framework against several general time series models, such as inverted Transformer (iTransformer) [61] and TimesNet [62]. We follow the methodologies provided by each corresponding paper during the experiment. More details refer to the supplementary.

D. Evaluation Protocols

Following the standard practice in blood pressure estimation [6], [9], we adopt common evaluation metrics, including root-mean-square error (RMSE), and Pearson's correlation coefficient values. RMSE is defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N-S} \sum_{i=1}^{N-S} (\hat{y}_i - y_i)^2}, \quad (15)$$

where $\hat{y}_i$ and y_i correspond to i -th estimated and true BP value (SBP or DBP), and $N-S$ is the total number of test samples. Pearson's correlation coefficient r is calculated to measure the linear correlation between the estimated and true BP values, defined as:

$$r = \frac{\sum_{i=1}^{N-S} (\hat{y}_i - \bar{Y})(y_i - \bar{Y}_{true})}{\sqrt{\sum_{i=1}^{N-S} (\hat{y}_i - \bar{Y})^2 \sum_{i=1}^{N-S} (y_i - \bar{Y}_{true})^2}}, \quad (16)$$

where $\bar{Y}$ and $\bar{Y}_{true}$ denote the mean values of the predicted ($\hat{y}$) and true BP values (y), respectively.

Additionally, we report the mean error (ME) and standard deviation of the error (SDE), as these metrics are recommended by the American National Standards Institute/ Association for the Advancement of Medical Instrumentation/ International Organization for Standardization AAMI standard. While the full AAMI protocol includes additional requirements such as minimum subject count and data distribution, thresholds of ME < 5 mmHg and SDE < 8 mmHg are commonly referenced in blood pressure estimation studies [63]. More results under the AAMI standard, please refer to Section V-J. Furthermore, we utilize a pair-wise t-test [64] to compare the performance of our proposed method and other baseline models, in order to make the experimental results more convincing. Detailed results of the pairwise t-tests are provided in the supplementary material.

TABLE I

RESULTS COMPARISON FOR BP ESTIMATION ON GRAPHENE-HGCPT DATASET IN TERMS OF OUR PROPOSED MODEL AND OTHER STATE-OF-THE-ART METHODS. *OURS-FULL* AND *OURS-BASE* REFER TO OUR FULL MODEL AND OUR MODEL WITHOUT ADVERSARIAL TRAINING AND CONTRASTIVE LEARNING, RESPECTIVELY. $\uparrow$ REFERS TO THE HIGHER RESULT BEING BETTER AND $\downarrow$ VICE VERSA. THE BOLD VALUES REPRESENT THE BEST PERFORMANCE AND UNDERLINED VALUES INDICATE THE SECOND BEST PERFORMANCE.

Models		iTransformer 2024		TimesNet 2023		ResNet1D 2023		Hybrid-LSTM 2022		PINN 2023		Ours-Base 2024		Ours-Full 2024	
BP type		Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$
SBP	1	0.41	16.5	0.38	11.5	0.18	13.3	0.35	20.1	<u>0.66</u>	9.5	0.63	<u>9.5</u>	0.68	9.0
	2	0.43	19.1	0.07	12.3	0.36	17.7	0.36	24.9	0.48	11.1	<u>0.61</u>	<u>9.6</u>	0.62	9.5
	3	0.48	17.0	0.40	12.9	0.16	15.5	0.42	16.8	0.43	12.6	<u>0.60</u>	<u>10.7</u>	0.61	10.6
	4	0.42	17.2	0.45	15.7	0.23	19.6	0.35	23.2	0.48	14.4	<u>0.57</u>	<u>13.0</u>	0.59	12.8
	5	0.27	18.6	0.30	13.6	0.28	17.7	0.20	22.3	0.47	12.1	<u>0.60</u>	<u>10.6</u>	0.62	10.6
	6	0.24	27.1	0.30	20.5	0.15	21.5	0.21	34.1	0.39	17.2	<u>0.59</u>	<u>14.9</u>	0.68	13.3
	Avg	0.38	19.3	0.32	14.4	0.23	17.6	0.32	23.6	0.48	12.8	<u>0.60</u>	<u>11.4</u>	0.63	11.0
DBP	1	0.43	16.5	0.23	10.8	0.49	12.3	0.42	18.3	<u>0.62</u>	8.3	0.62	<u>8.6</u>	0.61	8.7
	2	0.34	20.5	0.24	11.2	0.27	13.3	0.31	22.5	0.44	9.8	<u>0.54</u>	<u>9.1</u>	0.54	9.2
	3	0.44	15.2	0.38	11.4	0.21	10.3	0.32	16.6	0.51	9.8	<u>0.64</u>	<u>8.9</u>	0.66	8.6
	4	0.41	16.2	0.23	13.6	0.32	15.5	0.30	21.8	<u>0.50</u>	12.8	0.47	<u>11.9</u>	0.53	11.6
	5	0.26	20.0	0.33	15.8	0.33	19.6	0.17	26.3	0.60	11.9	<u>0.71</u>	<u>9.9</u>	0.74	9.4
	6	0.18	22.5	0.14	15.5	0.10	17.7	0.17	30.3	0.31	14.5	0.37	<u>13.5</u>	<u>0.34</u>	13.3
	Avg	0.34	18.4	0.26	13.1	0.29	21.5	0.28	22.6	0.50	11.1	<u>0.56</u>	<u>10.3</u>	0.57	10.1

E. Implementation Details

All implementations are based on the open-source PyTorch framework and were trained on a single NVIDIA 3090 GPU. The networks were optimized using the Adam optimizer with a learning rate of 0.0001. To generate controllable adversarial samples, we selected a relatively small perturbation parameter, $\epsilon = 0.2$, and set the number of training steps to 2. During training, we balanced the magnitudes of different loss components by adjusting hyperparameters based on a parameter sensitivity analysis. Specifically, we set the weight of the physics loss to $\gamma = 1$ and the minimum sensitivity of blood pressure to $y_{\text{shift}} = 2$. All models were trained with minimal labeled data. To ensure a fair comparison across all models, training was terminated once the supervised training loss reached 0.01 or reached the maximum training epochs, excluding the physics-based loss values from this calculation. The maximum number of training epochs was set to 2000. We provide a demonstration of training curves and testing curves in the supplementary material.

F. Results and Analysis

1) *Results on graphene-HGCPT Dataset:* Table I presents a comprehensive overview of the quantitative outcomes obtained through our proposed framework and alternative baseline approaches. From the table, it is evident that the Ours-Full model demonstrates superior performance against other methods (i.e., iTransformer, TimesNet, ResNet1D, Hybrid-LSTM) in both SBP and DBP estimation. For example, the Ours-Full model achieves a 31%/14% improvement in correlation and a 14%/9% reduction in RMSE (SBP/DBP) compared to the PINN model, mostly attributing to the superiority and effectiveness of our proposed temporal block for personal cardiovascular cycle modeling. Additionally, the improved results of the Ours-Full model compared to the Ours-Base model show that adversarial examples and contrastive learning are beneficial for improving blood pressure estimation.

Noting that under minimal training criteria, strong baselines (such as ResNet or Transformer-based methods) also struggle to make precise predictions and model BP dynamics, which is possibly due to their pure data-driven nature. Models with prior knowledge designed for time series can improve performance, thus TimesNet and PINN model achieves better results than iTransformer, ResNet1D, and Hybrid-LSTM. However, compared to our methods, TimesNet lacks physiological priors, while PINN models are insufficient for extracting temporal information. Our model combines physiological priors with advanced temporal modeling, enabling more precise blood pressure estimation.

Furthermore, the results underscore the varying challenges in modeling different subjects, as illustrated by the performance of subjects #4 and subject #6 in Table I, which show higher RMSE and lower correlation compared to other subjects. This highlights the critical importance of personalized modeling. Our model employs temporal blocks specifically designed for time series modeling, which better capture personalized cardiovascular cycles and enhance the accuracy of personal modeling.

2) *Results on ring-CPT Dataset:* The results and comparisons of our model with other state-of-the-art methods on the ring-CPT dataset can be found in Table II. From the table, we can see that the Ours-Full model manifests the best performance against other methods by a considerable margin. This improvement is largely attributed to our model's capability in temporal modeling and the effectiveness of our data augmentation methods, including adversarial examples and contrastive learning.

3) *Results on Blumio Dataset:* We report the results on the Blumio dataset to demonstrate our model is versatile and applicable to various modalities of cuffless blood pressure signals, such as mmWave and PPG. Table III presents the performance of our proposed approaches for SBP and DBP estimation using different modalities in the Blumio dataset (mmWave and PPG). From the table, it is evident that our proposed framework consistently achieves superior performance compared to other methods in terms of RMSE and Pearson's correlation. Specifically, when

TABLE II

RESULTS COMPARISON FOR BP ESTIMATION ON RING-CPT DATASET IN TERMS OF OUR PROPOSED MODEL AND OTHER STATE-OF-THE-ART METHODS. *OURS-FULL* AND *OURS-BASE* REFER TO OUR FULL MODEL AND OUR MODEL WITHOUT ADVERSARIAL TRAINING AND CONTRASTIVE LEARNING, RESPECTIVELY. $\uparrow$ REFERS TO THE HIGHER RESULT BEING BETTER AND $\downarrow$ VICE VERSA. THE BOLD VALUES REPRESENT THE BEST PERFORMANCE AND UNDERLINED VALUES INDICATE THE SECOND BEST PERFORMANCE.

Models	BP type	iTransformer 2024		TimesNet 2023		ResNet1D 2023		Hybrid-LSTM 2022		PINN 2023		Ours-Base 2024		Ours-Full 2024	
		Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$
SBP	1	0.62	10.5	0.67	8.8	0.58	10.7	0.50	14.2	0.58	9.0	<u>0.71</u>	<u>7.6</u>	0.73	7.4
	2	0.47	10.1	0.43	7.4	0.39	9.0	0.40	12.0	0.51	7.1	<u>0.66</u>	<u>6.2</u>	0.72	5.8
	3	0.43	14.9	0.42	11.3	0.60	11.2	0.48	17.1	0.65	9.8	<u>0.67</u>	<u>9.0</u>	0.69	8.8
	4	0.71	12.3	0.79	9.8	0.80	12.2	0.69	14.8	0.87	10.0	<u>0.90</u>	<u>6.5</u>	0.90	6.5
	5	0.54	8.8	0.50	8.1	0.56	8.8	0.49	11.5	0.68	7.4	<u>0.78</u>	<u>5.7</u>	0.82	5.2
	Avg	0.55	11.3	0.56	9.1	0.59	10.4	0.51	13.9	0.66	8.7	<u>0.74</u>	<u>7.0</u>	0.77	6.7
DBP	1	0.45	11.4	0.47	7.7	0.43	10.5	0.49	13.6	0.57	6.2	<u>0.66</u>	<u>6.0</u>	0.68	5.7
	2	0.36	9.4	0.20	6.8	0.38	5.8	0.21	12.9	0.46	<u>4.8</u>	<u>0.46</u>	5.5	0.53	4.8
	3	0.55	10.3	0.65	7.7	0.68	10.7	0.52	12.0	0.74	7.5	<u>0.78</u>	<u>6.4</u>	0.79	6.1
	4	0.69	10.4	0.62	8.7	0.76	12.9	0.67	12.2	0.81	6.4	<u>0.84</u>	<u>5.1</u>	0.84	5.1
	5	0.38	8.3	0.53	5.6	0.54	6.6	0.36	11.3	<u>0.72</u>	4.9	0.71	<u>4.6</u>	0.76	4.3
	Avg	0.49	10.0	0.49	7.3	0.56	9.3	0.45	12.4	0.67	6.0	<u>0.69</u>	<u>5.5</u>	0.72	5.2

TABLE III

RESULTS COMPARISON OF DIFFERENT MODAL (MMWAVE AND PPG) CUFFLESS SIGNAL BLOOD PRESSURE ESTIMATION ON BLUMIO DATASET. *OURS-FULL* AND *OURS-BASE* REFER TO OUR FULL MODEL AND OUR MODEL WITHOUT ADVERSARIAL TRAINING AND CONTRASTIVE LEARNING, RESPECTIVELY. $\uparrow$ REFERS TO THE HIGHER RESULT BEING BETTER AND $\downarrow$ VICE VERSA. THE BOLD VALUES REPRESENT THE BEST PERFORMANCE AND UNDERLINED VALUES INDICATE THE SECOND BEST PERFORMANCE.

Methods	mmWave		PPG	
	SBP/DBP		SBP/DBP	
	Corr $\uparrow$	RMSE $\downarrow$	Corr $\uparrow$	RMSE $\downarrow$
Hybrid-LSTM	0.27/0.26	7.0/6.1	0.24/0.27	7.9/7.3
ResNet1D	0.29/0.30	6.7/6.0	0.28/0.35	7.2/5.9
iTransformer	0.27/0.27	6.6/5.2	0.28/0.32	7.5/5.7
TimesNet	0.22/0.28	4.1/3.6	0.22/0.27	4.3/3.7
PINN	0.45/0.37	<u>3.8/3.5</u>	0.41/ <u>0.40</u>	<u>4.0/3.7</u>
Ours-Base	0.47/0.40	3.8/3.4	0.42/0.40	4.2/3.8
Ours-Full	0.51/0.44	3.7/3.3	0.44/0.42	3.5/3.7

using mmWave for BP estimation, the Ours-Full model achieves a 132%/57% improvement in correlation (SBP/DBP) compared to the TimesNet. Similarly, for BP estimation using PPG, the Ours-Full model shows a 100%/56% improvement in correlation (SBP/DBP) over the TimesNet. The enhanced performance of the Ours-Full model on Pearson's correlation is mostly due to the physiological prior and contrastive learning. However, the correlation results achieved in the Blumio using either PPG or mmWave modalities are generally inferior to those in the graphene-HGCPT and ring-CPT datasets, which may stem from the sparse sampling points, causing inconsistent predictions.

4) *Qualitative Results Analysis*: Fig. 4 visually compares the beat-to-beat SBP and DBP estimations using the Ours-Full model and PINN model. The results clearly demonstrate the effectiveness of the proposed model. Under identical training constraints and experimental conditions, our findings show that the Ours-Full model surpasses the PINN model in representing subject-specific characteristics, yielding superior correlation and reduced absolute error. To facilitate reproducibility, Fig. 4 presents a demo subset from ring-CPT subject #5, and the full reproduction pipeline is provided in our GitHub repository.

Ours-Full model consistently outperforms the PINN model across all three signal types (bioimpedance, mmWave, and PPG). Cardiovascular time series data typically involve intricate temporal patterns, where multiple variations (e.g., rising, falling, fluctuation) mix and overlap, making temporal variation modeling extremely challenging. Compared to the vanilla PINN, our PITN model is better equipped to capture such temporal information, resulting in a closer fit to the actual blood pressure values. However, there are still a few failure cases, such as subject #VR69 with poor performance. We believe this could be attributed to the presence of transient variations in BP, which were not well captured during training. These abrupt, short-term fluctuations in BP dynamics, which are not consistent across subjects, present a challenge for the model in predicting BP accurately.

As a proof-of-concept, we present Pearson's correlation visual analysis in Fig. 5, by using subject #5 from the graphene-HGCPT as an example. We can clearly observe that the prediction points of the Ours-Full model are more closely aligned with the dashed line (where predicted values equal true values), while those of the PINN model are more dispersed. This further demonstrates the effectiveness of our proposed methods in personal modeling and in capturing discriminative variations and cardiovascular dynamics.

G. Efficiency Analysis

Training PITN requires approximately 0.5 s per epoch on an RTX 3090 GPU. Additionally, to assess the feasibility of CPU-side inference, we successfully loaded a pre-trained PITN model on an Intel(R) Core(TM) i7-10750H processor. Given the real-time requirements of continuous blood pressure (BP) estimation, we conducted experiments to evaluate inference time. Our proposed PITN model takes 0.02 s to infer a single segment, and the number of multiply-accumulate (MAC) operations in PITN is approximately 150 M. Inference time is calculated on a single RTX 3090. For a preprocessing time, each segment takes about 0.015 s. The preprocessing is conducted on an Intel(R) Core(TM) i7-10750H CPU. Modern wearable

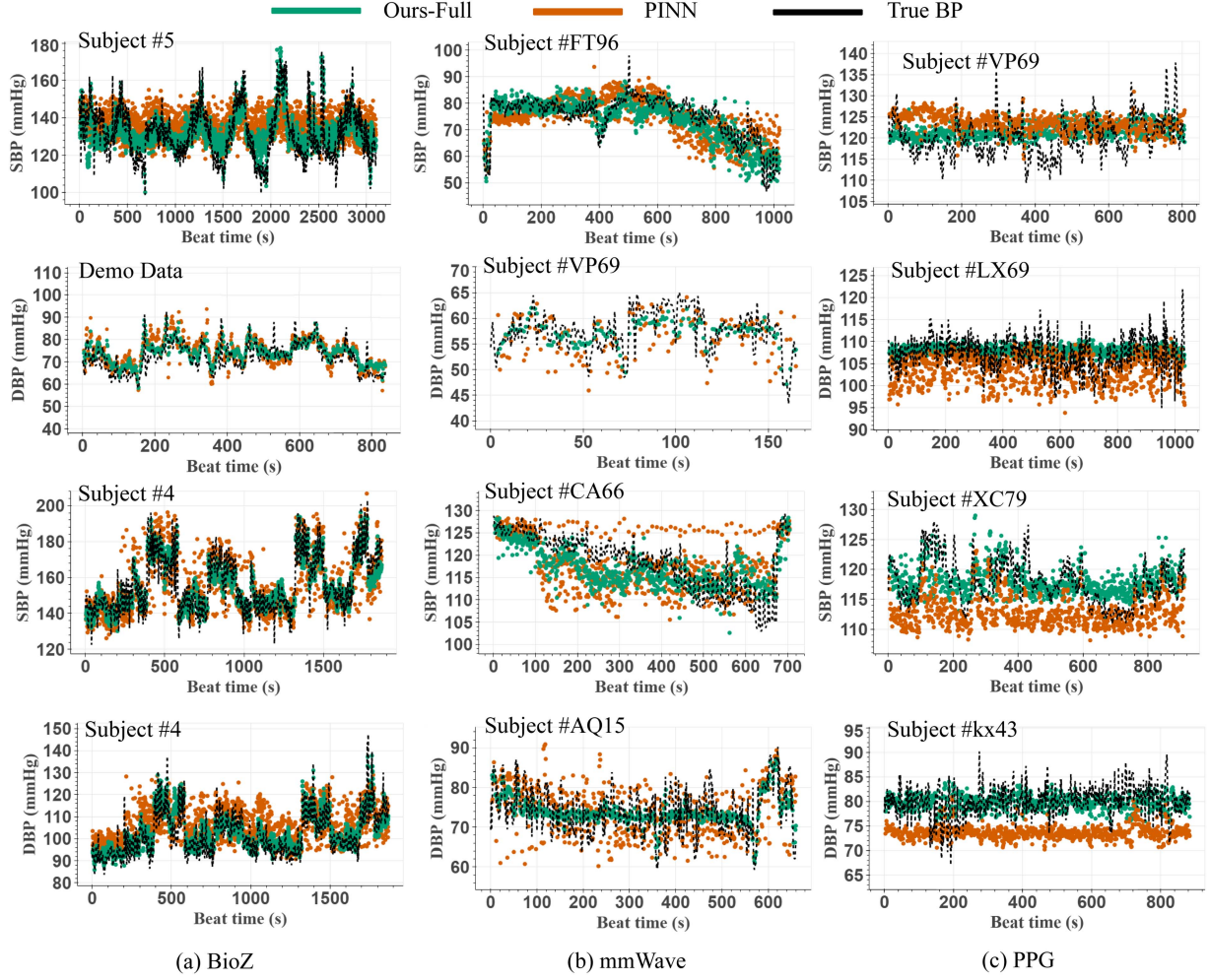


Fig. 4. Beat-to-beat SBP and DBP estimation based on Ours-Full model (shown in green) and PINN (shown in orange) model trained with the same number of instances and corresponding true BP (shown in dashed black). The figures in each column correspond to the same signal, with the subject listed at the top of each figure. Ours-Full model shows a more precise fit to the reference BP.

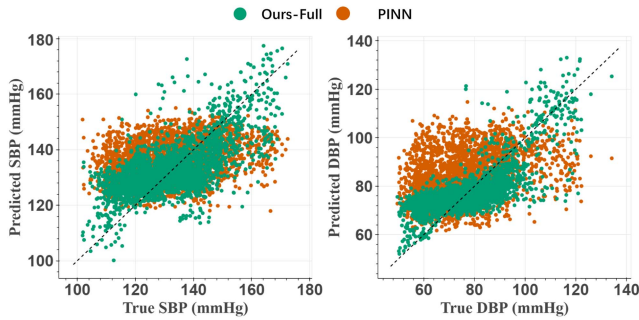


Fig. 5. Pearson's correlation analysis with 2910 blood pressure values using subject #5 in the graphene-HGCPT dataset. Predictions of the Ours-Full model are in green, while predictions of the PINN model are in orange.

devices, such as the Apple Watch (S9 chip) [65] and Huawei Watch D2 [66], incorporate efficient neural network accelerators capable of running lightweight deep learning models. Given that our model demonstrates lower MACs compared to KD-Informer (300 M) [16], this suggests that it could be efficiently executed on embedded hardware with sufficient optimization.

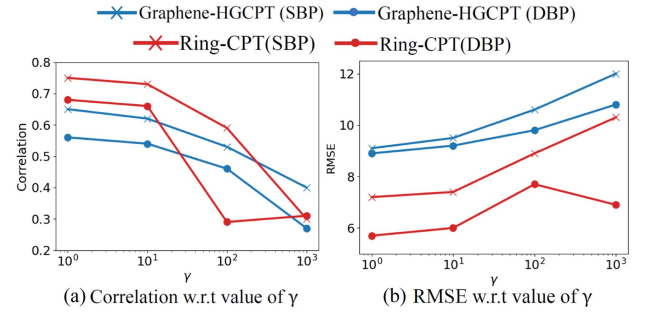


Fig. 6. Parameter sensitivity analysis of value of the weight of physics loss γ w.r.t. correlation and RMSE on the graphene-HGCPT and ring-CPT datasets.

H. Parameter Sensitivity and Convergence Analysis

Here we assess the impact of the weight of physics loss γ when training our model with a randomly selected subset of the two datasets. As shown in Fig. 6, magnitude of physics loss $\mathcal{L}_{\text{physics}}$ are relatively not consistent with loss of regression $\mathcal{L}_{\text{MSE}}$, thus we ablate the weight of physics loss γ in [1, 10, 100, 1000] for balancing the magnitude of multiple losses. Fig. 6 shows

TABLE IV

ABLATION STUDY ON PROPOSED METHODS. *OURS-BASE*, *OURS w/ ADV*, AND *OURS-FULL*, DENOTE THE BASELINE MODEL, OUR MODEL ONLY WITH ADVERSARIAL SAMPLES, AND OUR FULL MODEL, RESPECTIVELY. THE PERFORMANCE IMPROVEMENT IS COMPARED TO THE PINN MODEL. $\uparrow$ REFERS TO THE HIGHER RESULT BEING BETTER AND $\downarrow$ VICE VERSA. THE BOLD VALUES REPRESENT THE BEST PERFORMANCE AND UNDERLINED VALUES INDICATE THE SECOND BEST PERFORMANCE.

Model Name	Method			Graphene-HGCPT				Ring-CPT			
	Temporal block	Adversarial	Contrastive	Corr $\uparrow$		RMSE $\downarrow$		Corr $\uparrow$		RMSE $\downarrow$	
				SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
PINN [6]	$\times$	$\times$	$\times$	0.48	0.50	12.8	11.1	0.66	0.67	8.7	6.0
Ours-Base	$\checkmark$	$\times$	$\times$	+0.12	+0.06	-1.4	-0.8	+0.08	+0.02	-1.7	-0.5
Ours w/ adv	$\checkmark$	$\checkmark$	$\times$	<u>+0.13</u>	+0.05	<u>-1.7</u>	<u>-0.9</u>	<u>+0.09</u>	<u>+0.03</u>	<u>-1.9</u>	-0.8
Ours-Full	$\checkmark$	$\checkmark$	$\checkmark$	+0.15	+0.07	-1.8	-1.0	+0.11	+0.05	-2.0	-0.8

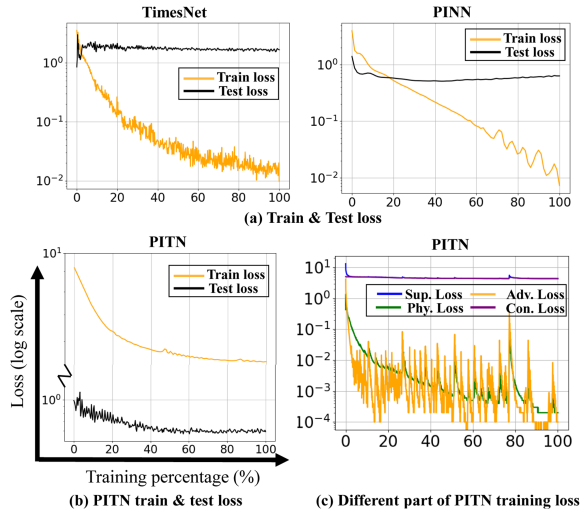


Fig. 7. Training/testing loss curves for baseline models and PITN, and PITN loss components.

how correlation and RMSE change according to the weight of physics loss γ . It is evident that as the value of γ increases, the correlation decreases while the RMSE increases, indicating a decline in performance. Through Fig. 6, we can determine that employing a γ value of 1 yields the best accuracy.

As shown in Fig. 7(a), (b), we present the training and testing loss curves for TimesNet, PINN, and PITN, which clearly illustrate the convergence behavior and potential overfitting trends. PITN demonstrates consistently stable convergence, resulting in superior generalization performance, while PINN model exhibits a slight test loss increase. Fig. 7 illustrates all training loss components in PITN. Overall, the losses exhibit proper convergence behavior, with physics and adversarial losses maintaining low values and contrastive loss remaining stable. The observed oscillations in adversarial loss are likely due to sharp loss landscapes near adversarial samples.

I. Ablation Study

We conduct ablation studies to examine the effectiveness of the proposed temporal block, adversarial training, and contrastive learning, respectively.

1) *Temporal Block*: As shown in Table IV, with temporal modeling, the Ours-Base model outperforms the vanilla PINN

model by a large margin. From the table, we can observe that our proposed temporal blocks have a significant influence on BP prediction, removing them will lead to a significant drop in performance. This demonstrates the vital significance of the proposed temporal block. In terms of Pearson correlation, Ours-Base excels the vanilla PINN by relatively 25% and 12% in SBP and DBP estimation, while in terms of RMSE, Ours-Base outperforms the vanilla PINN by 11% and 3% relatively in SBP and DBP estimation.

2) *Adversarial Training*: As shown in Table IV, we can see augmenting the framework with adversarial samples (denoted as “Ours w/ adv”) enhances performance compared to the Ours-Base model, which utilizes only clean samples for training. This improvement suggests that minor perturbations in adversarial samples can enhance model performance, especially when training data is insufficient. Although these adversarial samples are designed to introduce perturbations through negative gradients, careful control of the generation steps and perturbations allows them to supplement the training data rather than disrupt it. Furthermore, we observe that for adversarial samples, the Pearson’s correlation between the predicted and true BP values slightly increases (except for DBP estimation in the graphene-HGCPT dataset). This suggests that the inclusion of adversarial samples helps mitigate overfitting to some extent.

For a clear demonstration of the difference and similarity between adversarial samples and clean samples, we provide waveform and T-SNE clustering visualization in Fig. 8. The signal used is BioZ. It is evident that, although the generated adversarial examples differ in signal characteristics from the clean examples in terms of the waveform, their primary feature components remain closely aligned with the corresponding clean examples, as shown in the T-SNE visualization. This finding indicates that minor perturbations provide valuable learning opportunities for the model by introducing controlled disturbances, without altering the signal’s original semantic content. The perturbation introduced is substantial yet non-detrimental, demonstrating the effectiveness of our adversarial training approach. Additionally, we compare various data augmentation methods in the supplementary material to further demonstrate the superiority of our approach.

3) *Contrastive Learning*: As shown in Table IV, our full model achieves the best correlation between state-of-the-art methods and the Ours-Base model, demonstrating the effectiveness of incorporating contrastive learning to capture BP flow

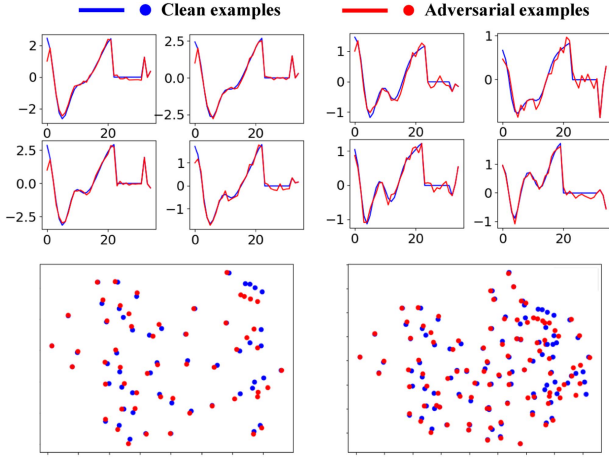


Fig. 8. Visualizations of adversarial and clean examples using processed BioZ signals from the graphene-HGCPT dataset. The top row illustrates waveform representations, while the bottom row presents T-SNE projections for a comparative analysis.

TABLE V
BLOOD PRESSURE ESTIMATION PERFORMANCE AGAINST AAMI STATISTICAL REQUIREMENTS (MEAN ERROR $\pm$ STANDARD DEVIATION)

Ours-Full	SBP		DBP	
	ME (mmHg) < 5	SDE (mmHg) < 8	ME (mmHg) < 5	SDE (mmHg) < 8
Dataset (Signal type)	ME (mmHg)	SDE (mmHg)	ME (mmHg)	SDE (mmHg)
Graphene-HGCPT	1.5	8.3	1.8	7.9
Ring-CPT	-0.5	6.8	1.2	5.0
Blumio (PPG)	-0.7	5.0	-1.3	7.3
Blumio (mmWave)	-0.5	4.1	0.3	3.4

changes over time. Contrastive learning can enhance model performance, provided that proper positive-negative sample pairs are available. We construct these positive-negative pairs through BP values, where waveforms with similar BP labels are treated as positive pair, while those with dissimilar BP values are considered negative pairs. As anticipated, by aligning bioimpedance signals with similar BP values, our framework can better capture BP dynamics, resulting in improved Pearson's correlation coefficients. Furthermore, by precisely modeling personal discriminative variations, our full model obtained slightly better performance in RMSE compared to the base model. Overall, it can be observed that, compared to the Ours-Base model, our full model effectively learns cardiovascular dynamics and personal discriminative variations.

J. Discussion

Systolic and diastolic blood pressure values correspond to the maximum and minimum pressure points in the artery. During systole, the heart ejects blood into the aorta, which then travels through the arterial tree. Changes in SBP and DBP are proportional to volumetric changes based on the arterial wall's characteristics, which are defined by compliance. Since arterial compliance varies across subjects, precise personal modeling is essential. We evaluate our model against the core statistical criteria of AAMI (Association for the Advancement of Medical Instrumentation) standards [63]. As shown in Table V, our proposed methods meet the AAMI's standard in terms of

ME and STD across graphene-HGCPT, ring-CPT, and Blumio (mmWave and PPG) datasets, with the exception of SBP estimation in the Graphene-HGCPT dataset. Although the proposed model achieves an SDE less than 8 mmHg for estimating SBP and DBP in most cases, its performance in SBP estimation was relatively lower, possibly due to individual variations in higher SBP values. These findings are consistent with previous studies [67], [68], [69], which have highlighted the need to enhance the performance of cuffless BP models, particularly for populations with higher or hypertensive BPs. Additionally, to enhance the precision of BP estimation, incorporating multiple sensors can be promising for future research. By integrating various modalities of cuffless sensor signals, the accuracy of BP estimation could be improved.

VI. CONCLUSION

In this paper, we presented an adversarial contrastive learning-based Physics-Informed Temporal Network for cuffless blood pressure estimation. Specifically, we introduced a temporal block within physics-informed neural networks to extract intra-periodic physiological features for temporal modeling. Additionally, we implemented adversarial training combined with contrastive learning to augment physiological time series data. Extensive experiments conducted on multimodal signals from the graphene-HGCPT, ring-CPT, and Blumio datasets, have demonstrated the superior effectiveness of our framework. In the future, we will apply the proposed framework to other medical applications, such as blood sugar monitoring.

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